Quantitative Research

Part 1: Descriptive Statistics

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About the Lecturer



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MLitt (St Andrews), MSc (Oxford)

- Professor of Mathematics, specialising in Business Data Science
- Former Technical Lead for Artificial Intelligence and supervisor of industrial doctoral projects at BASF
- **Research:** Data Science, Machine Learning and Artificial Intelligence
- **Focus:** application of AI in business and industry

Course materials:
github.com/ChrisW09/Quantitative-Research-Methods

The lecture at a glance

	Part	Content
▶	**1 – Descriptive Statistics**	**Basic concepts, levels of measurement, frequencies, measures of central tendency and dispersion**
	2 – Graphical Representations	Empirical CDF, histogram, boxplot
	3 – Statistical Inference	Normal distribution, CLT, confidence intervals, sample size, hypothesis tests, p-values, one-sided tests, types of error
	4 – Regression	Correlation, least squares, goodness of fit, inference for regression, multiple regression (appendix: residual analysis)
	5 – Classification	Conditional probability, logistic regression, classification rules, confusion matrix, ROC/AUC, Bayes classifiers
	Python course	Separate course: Python basics, pandas, plots, machine learning (Colab notebooks)

All analyses in **Python** (NumPy, pandas, SciPy, statsmodels, matplotlib); every deck comes with a Python script and a Jupyter notebook – also in Google Colab.

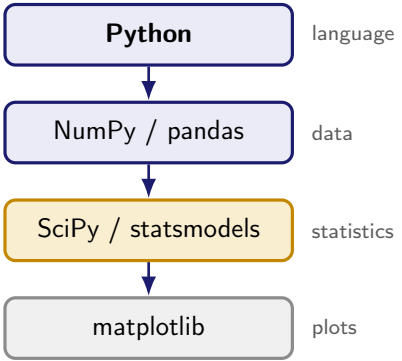
Software and References: Python

W. McKinney. *Python for Data Analysis*. 3rd ed., O'Reilly, 2022. Free online: `wesmckinney.com/book`

J. VanderPlas. *Python Data Science Handbook*. 2nd ed., O'Reilly, 2023.

All software used in this course is **free and open source**.

No Python yet? The separate **Python course** (folder `Python_Course`) starts from zero – all notebooks run in Google Colab.



▶ **New to Python?** Separate Python course: Track 1, a 90-minute introduction – open on GitHub

Contents – Part 1

1. Fundamentals
2. Descriptive Statistics
3. Summary

- **Try it yourself:** notebook, all examples and exercises of this part – open in Colab

Learning objectives – Part 1

After this part you will be able to ...

- name the **basic concepts** (statistical unit, variable, value, observation) and the **levels of measurement**,
- compute **absolute, relative and cumulative frequencies** and read off shares ("at most", "at least"),
- compute and interpret **measures of central tendency** (means, median, mode, quantiles),
- average **growth rates** with the geometric mean,
- compute and compare **measures of dispersion** (R, IQR, s^2, s, v, v^*),
- choose the **right measure** for a given comparison,
- carry out every computation in **Python**.

Key takeaway

Running story: the **shoe sizes of a soccer team** – in numbers (Part 1), in pictures (Part 2). **Exercises** follow every topic; an **exam-style exercise** closes the part.

List of Symbols (1)

Symbol	Meaning	Symbol	Meaning
ω	Statistical unit	h_j	Absolute frequency
n	Number of statistical units	f_j	Relative frequency
X	Variable	H_j	Cumulative absolute frequency
x	Value (category)	$F(x)$	Empirical distribution function
x_i	Observation	x_j	Value (class j)
$x_{(i)}$	i-th value of the ordered sample	k	Number of distinct values
$\sum$	Sum (summation sign)	$\prod$	Product (product sign)

List of Symbols (2)

Symbol	Meaning	Symbol	Meaning
$\bar{x}$	Arithmetic mean	s^2	Variance (sample)
$\bar{x}_{\text{trim}}$	Trimmed mean	s	Standard deviation
$\bar{x}_{\text{geo}}$	Geometric mean	R	Range
r_i	Growth rate in period i	IQR	Interquartile range
$1 + r_i$	Growth factor in period i	v	Coefficient of variation
$\tilde{x}$	Median	v^*	Normalised coeff. of variation
x_p	p-quantile	$\lceil\cdot\rceil$, $\lfloor\cdot\rfloor$	Rounding up / down to an integer
Q_1, Q_3	1st and 3rd quartile	$x_{0.25}$, $x_{0.75}$	25 %- and 75 %-quantile

Symbols for regression (s_{xy}, r, β_0, β_1, R^2, ...) are introduced in Part 4.

Overview

1. Fundamentals
   - Basic Concepts
   - Levels of Measurement
   - Exercise
2. Descriptive Statistics
3. Summary

Basic Concepts

Statistical unit (ω): Individual object being observed. Must be delimited in terms of subject matter, time and location, e.g. a particular day at a particular weather station (variable: the daily maximum temperature).

n: The number of statistical units

Sample: The collected set of statistical units

Population: The overarching set from which the sample was drawn

Basic Concepts: Variable and Observed Values

Variable (X): A characteristic of the statistical unit. There may be several characteristics; which one is examined must be specified.

Values (x): Possible values that the variable can take.

Observation (x_i): The value that was actually observed for the variable at the i-th statistical unit.

Basic Concepts – Summary

Term	Symbol	Explanation	Example
Statistical unit	ω	Unit of analysis	Persons, institutions, objects
Variable	X	Characteristic of interest of a statistical unit	Age, market value, weight, height, costs, rating, product count, colour
Value	x	Possible values, states or categories that a variable can take	18, 19, 20, ... years; grades 1 to 5; yes or no; black, white, blue, ...
Observation	x_i	The value actually observed at the i-th statistical unit	31 years, 1300 euros, 23 cm, 73 kg, black

Running example of this part

ω = a player ($n = 11$), X = shoe size, values x: 37, 38, ..., observation of player 1: $x_1 = 40$

Data Collection and Survey Types

Data Collection

- **Primary data:** Data collected specifically for this study
- **Secondary data:** Data taken from existing statistics

Survey Types

- Observation
- Interview / Questionnaire
- Experiment

Data Protection

- Regulated by law, e.g. the GDPR
- Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data, on the free movement of such data, and repealing Directive 95/46/EC
- As well as further national legislation

Important

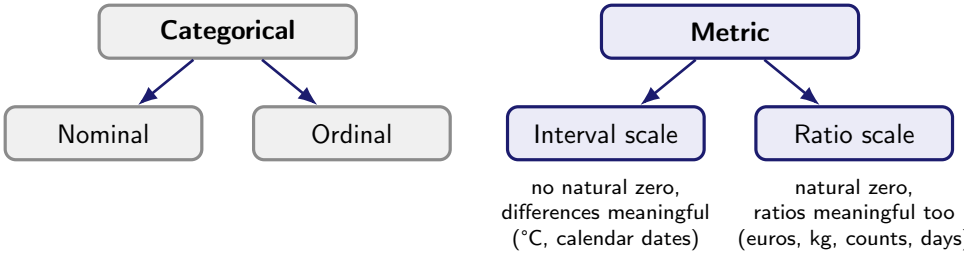
Compliance with data protection is a prerequisite for data analysis!

Levels of Measurement (1)

Term	Explanation	Meaningful operations	Example
Nominal scale	Category only, no ordering	$=, \neq$	Colour, gender, origin, occupation
Ordinal scale	Ranking possible, differences not meaningful	additionally $<, >$	Attitudes, well-being, grades
Metric scale	Ranking possible, differences meaningful	additionally $+, -$	Monetary units, temperatures

In words: Each level allows everything the level above it in the table allows – and one thing more. The higher the level, the more we may compute with the data.

Levels of Measurement (2)



- Categorical data includes nominal and ordinal data.
- The metric level splits into the interval and the ratio scale: "twice as expensive" makes sense for prices, "twice as warm" does not for °C.

Important

A metric level of measurement is a prerequisite for most of the statistical analyses covered in this course!

Levels of Measurement with Python

```python
import pandas as pd
# Nominal: categories without an order -> data type "category"
colour = pd.Series(["black", "white", "blue", "black"], dtype="category")
print((colour == "black").tolist())     # [True, False, False, True]   (=, != only)
# Ordinal: categories WITH an order (clothing sizes S < M < L < XL)
size = pd.Series(pd.Categorical(["M", "S", "XL", "M"],
                 categories=["S", "M", "L", "XL"], ordered=True))
print(size.max(), (size > "S").sum())  # XL 3                          (additionally <, >)
# Metric: numbers -> differences (and means) are meaningful
price = pd.Series([19.90, 24.90, 29.90, 19.90])   # euros
print(round(price.max() - price.min(), 2), round(price.mean(), 2))   # 10.0 23.65
```

Pitfall: numbers used as codes

pandas treats every number column as metric. Codes such as customer IDs must be converted with `.astype("category")` – otherwise `.mean()` returns a meaningless value.

▶ **Try it yourself:** notebook, section 1.2 "Levels of Measurement" – open in Colab

Exercise – Levels of Measurement

Problem statement

An online retailer records the following variables for every order (statistical unit):

	Variable		Variable
1	Postcode of the delivery address	5	Order value in euros
2	Payment method (invoice, PayPal, card)	6	Delivery time in days
3	Customer rating (1 to 5 stars)	7	Customer status (new / regular / premium)
4	Number of items ordered	8	Order date (calendar date)

a) Assign each variable its level of measurement (nominal, ordinal, metric); give a brief reason.

b) For the metric variables: interval or ratio scale? (Is "twice as much" meaningful?)

Solution – Levels of Measurement

Rule: nominal → categories; ordinal → + ranking; metric → + differences; ratio → + natural zero.

	Variable	Level	Reason
1	Postcode	**nominal**	the number is only a code, no ranking
2	Payment method	**nominal**	categories without an order
3	Customer rating	**ordinal**	clear ranking, but distances between stars not comparable
4	Number of items	**metric (ratio)**	differences (3 items more) and ratios meaningful
5	Order value	**metric (ratio)**	natural zero (0 euros); "twice as expensive" meaningful
6	Delivery time	**metric (ratio)**	differences (2 days longer) and ratios meaningful
7	Customer status	**ordinal**	new < regular < premium, but no measurable distance
8	Order date	**metric (interval)**	distances in days meaningful, but no natural zero

Remember

Numbers used as codes say nothing about the level of measurement: postcodes are nominal – an "average postcode" is meaningless.

Overview

1 Fundamentals

2 Descriptive Statistics
- Frequencies
- Exercises
- Measures of Central Tendency
- Exercises
- Measures of Dispersion
- Exercises
- Exam-Style Exercise

3 Summary

Frequencies

Frequencies – Absolute, Relative, Cumulative

Frequencies

For each distinct value x_j $(j = 1, \ldots, k)$ of the variable, sorted in ascending order:

Frequency	**Formula**	**Answers the question ...**
Absolute	$h_j = \#\{i \mid x_i = x_j\}$	How often does the value x_j occur?
Relative	$f_j = \dfrac{h_j}{n}$	Which share of the observations equals x_j?
Cumulative absolute	$H_j = \sum_{l \le j} h_l$	How many observations are $\le x_j$?
Cumulative relative	$F(x) = \sum_{x_j \le x} f_j$	Which share of the observations is $\le x$?

In words: Count, divide by n, add up from the left. Checks: $\sum_j h_j = n$ and $\sum_j f_j = 1$; the last cumulative values are $H_k = n$ and $F = 1$.

The Data – an Example (1)

The statistical units: the players of a soccer team



The Data – an Example (2)

The variable: The shoe sizes of a soccer team with the following observations:

40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39

Frequency Table

Shoe size	37	38	39	40	41	46	Total
Absolute frequency	1	2	4	2	1	1	11
Relative frequency (decimal)	0.0909	0.1818	0.3636	0.1818	0.0909	0.0909	1
Relative frequency in %	9.1 %	18.2 %	36.4 %	18.2 %	9.1 %	9.1 %	100 %
Relative frequency (fraction)	$\frac{1}{11}$	$\frac{2}{11}$	$\frac{4}{11}$	$\frac{2}{11}$	$\frac{1}{11}$	$\frac{1}{11}$	1
Cumulative absolute frequency	1	3	7	9	10	11	
Cumulative rel. freq. (decimal)	0.0909	0.2727	0.6364	0.8182	0.9091	1	
Cumulative rel. freq. (fraction)	$\frac{1}{11}$	$\frac{3}{11}$	$\frac{7}{11}$	$\frac{9}{11}$	$\frac{10}{11}$	1	

Comparing Relative Frequencies: Percent vs. Percentage Points

Example: In a customer survey, the share of 5-star ratings rises from $f = 40\,\%$ last year to $f = 50\,\%$ this year.

Absolute difference of two shares

$$50\,\% - 40\,\% = \textbf{10 percentage points}$$

Relative change of the share

$$\frac{50\,\% - 40\,\%}{40\,\%} = 0.25 = \textbf{25}\,\%$$

In words: "The share rose by 10 percentage points" and "the share rose by 25 %" describe the *same* change – once as a difference, once relative to the starting level.

Key takeaway

Percentage points = **absolute** difference of two percentages; percent = **relative** change. The two are often confused in business news!

Exercise – Frequencies of a Nominal Variable

Problem statement

The online retailer from the exercise on levels of measurement records the payment method of $n = 10$ orders (I = invoice, P = PayPal, C = card):

P, C, I, P, P, C, P, I, C, P

a) Determine the absolute frequencies h_j and the relative frequencies f_j (fraction, decimal, %).

b) Which share of the orders is **not** paid by invoice? How many orders is that?

c) Last month, 25 % of the orders were paid by invoice. By how many percentage points and by how many percent has this share changed?

d) Does a cumulative relative frequency $F(x)$ make sense here? Give a reason.

Solution (1/2) – Frequencies of a Nominal Variable

Rule

h_j = number of values equal to x_j; $f_j = h_j/n$; checks: $\sum_j h_j = n$ and $\sum_j f_j = 1$.

a) Count the $n = 10$ orders (I: 2 times, P: 5 times, C: 3 times); column "Total" = check:

Payment method x_j	Invoice	PayPal	Card	Total
Absolute frequency h_j	2	5	3	10
Relative frequency f_j (fraction)	$\frac{2}{10}$	$\frac{5}{10}$	$\frac{3}{10}$	1
Relative frequency f_j (decimal)	**0.20**	**0.50**	**0.30**	1
Relative frequency in %	20 %	50 %	30 %	100 %

b) "Not by invoice" = PayPal or card: $f_{\text{PayPal}} + f_{\text{card}} = 0.50 + 0.30 = \mathbf{0.80}$ $(= 1 - f_{\text{invoice}})$, i.e. $0.80 \cdot 10 = \mathbf{8}$ orders (check: $h_{\text{PayPal}} + h_{\text{card}} = 5 + 3 = 8$).

Solution (2/2) – Frequencies of a Nominal Variable

c) Share paid by invoice: last month 25 %, now $f_{\text{invoice}} = 20\,\%$:

- absolute difference: $20\,\% - 25\,\% = -\mathbf{5}$ **percentage points**
- relative change: $(20\,\% - 25\,\%)/25\,\% = -0.20 = -\mathbf{20}\,\%$

d) No: the payment method is **nominal** – there is no ordering, so "$x_j \leq x$" is not defined. A cumulative value "up to PayPal" would depend on the arbitrary order of the columns:

- order invoice, PayPal, card: $0.20 + 0.50 = 0.70$
- order card, PayPal, invoice: $0.30 + 0.50 = 0.80$

Remember

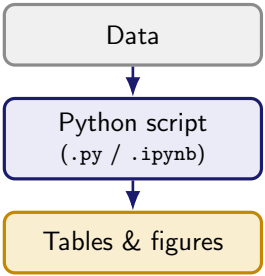
h_j and f_j exist for every level of measurement, $F(x)$ only from the **ordinal** level upwards.

Why Statistics Software?

Advantages of Modern Software

- Complex calculations in seconds
- Reproducible analyses through scripts
- Professional visualisations
- Ability to handle large datasets

In this course: We use **Python** with **Jupyter** notebooks or **VS Code**



www.python.org

What is Python?

- **Python** is a programming language for statistics, data analysis and much more.
- Typical application areas:
  - Data Science
  - Statistics
  - Machine Learning
  - Scientific research
- Strengths of Python:
  - Easy-to-read syntax, huge community, free and open source
  - Large selection of packages (`numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`)
  - Same language from data analysis to machine learning and production
- **Important:** Python + packages are the *computation engine* (perform calculations).

▶ **New to Python?** Separate Python course: notebook "Welcome to Python" – open on GitHub

Working Environment: Jupyter and VS Code

- **Jupyter notebooks** (`.ipynb`) and **VS Code** are the working environments (IDEs) for Python.
- They provide a user-friendly interface:
  - Jupyter: code cells with tables and plots directly below, plus text for explanations
  - VS Code: editor for scripts (`.py`) and notebooks, integrated terminal
  - Plot windows, variable explorer, package management
- **Analogy:**
  - Python = Engine of a car
  - Jupyter / VS Code = Dashboard and steering wheel
- **Important:** Jupyter and VS Code only work if Python is installed.

Installation (Mac & Windows)

- **Step 1: Install Python**
  - Recommended: **Anaconda Distribution** (`https://www.anaconda.com/download`) – bundles Python, all packages and Jupyter
  - Alternative: Python from `https://www.python.org`, then in a terminal:
    `pip install numpy pandas scipy statsmodels matplotlib jupyter`
- **Step 2: Choose a working environment**
  - Jupyter: start **JupyterLab** (included in Anaconda) or run `jupyter lab`
  - VS Code: `https://code.visualstudio.com` + extensions "Python" and "Jupyter"
- **Step 3: First test** – run in a notebook cell; no error message means you are ready:

```
import numpy, pandas, scipy, statsmodels, matplotlib
print(numpy.__version__, pandas.__version__)
```

▶ **Try it yourself:** notebook, setup cell (Colab needs no installation) – open in Colab

Frequencies with Python

```python
import pandas as pd

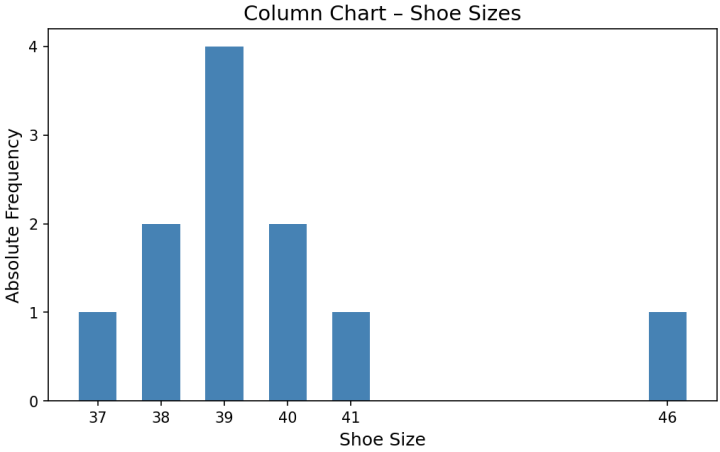
# Soccer team - shoe sizes of the 11 players
data = pd.Series([40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39])

# Frequency table (absolute frequencies), sorted by shoe size
print(data.value_counts().sort_index())
# Relative frequencies
print(data.value_counts(normalize=True).sort_index())
# Cumulative (relative) frequencies
print(data.value_counts(normalize=True).sort_index().cumsum())
# Percentage points vs. percent (share of 5-star ratings: 40 % -> 50 %)
print(50 - 40, round((50 - 40) / 40, 4))      # 10 0.25
```

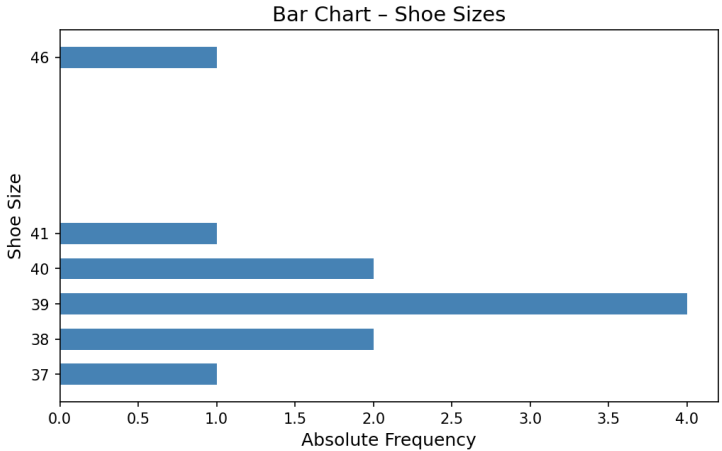
Note: `value_counts()` sorts by frequency; `sort_index()` orders by shoe size (needed for `cumsum()`).

▶ **Try it yourself:** notebook, section 2.1 "Frequencies" – open in Colab

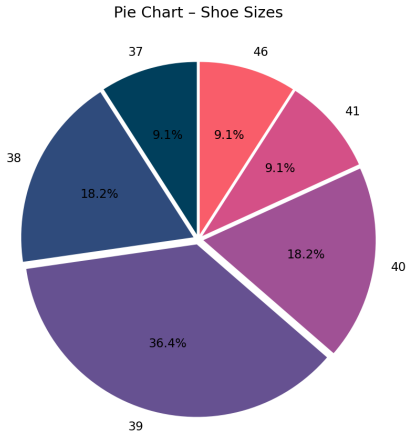
Column Chart



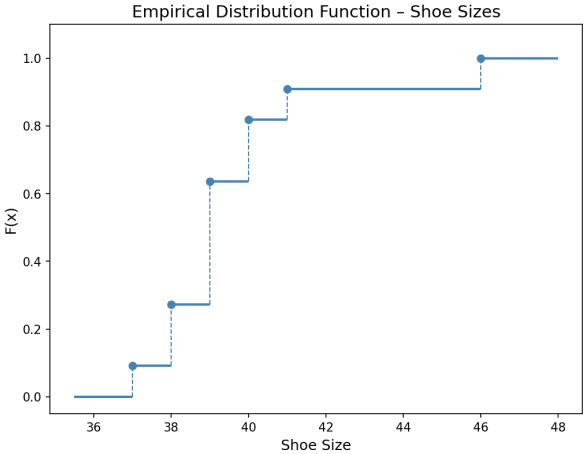
Bar Chart


Bar Chart – Shoe Sizes
Shoe Size
46
41
40
39
38
37
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Absolute Frequency

Pie Chart



Empirical Distribution Function (Empirical CDF, ECDF)



Charts with Python

```
import matplotlib.pyplot as plt
counts = data.value_counts().sort_index()        # h_j: 37:1, 38:2, 39:4, 40:2, 41:1, 46:1

plt.bar(counts.index, counts.values)             # column chart (numeric axis: gap 41-46)
plt.xlabel("Shoe size"); plt.ylabel("Absolute frequency")
plt.show()
plt.barh(counts.index, counts.values)            # bar chart (horizontal bars)
plt.show()
plt.pie(counts.values, labels=counts.index)      # pie chart: slice = share f_j
plt.show()
plt.ecdf(data)                                   # empirical distribution function F(x)
plt.show()
```

Note: `plt.ecdf()` needs matplotlib 3.8 or newer (included in Anaconda and Colab); `plt.show()` displays each chart below the cell.

▶ **Try it yourself:** notebook, section 2.1 "Frequencies" – open in Colab

Exercise – Frequency Table

Problem statement

A café has received $n = 12$ ratings (1 to 5 stars) on a review platform:

$$4,\ 5,\ 3,\ 4,\ 5,\ 5,\ 2,\ 4,\ 5,\ 3,\ 4,\ 5$$

a) Set up the frequency table with absolute frequencies h_j, relative frequencies f_j (as fraction and decimal) and cumulative relative frequencies $F(x_j)$.

b) What share of the ratings has **at most** 3 stars?

c) What share of the ratings has **at least** 4 stars? How many ratings is that?

d) Read from the table: which rating is the most frequent one?

e) Last year, 30 % of the ratings had 5 stars. By how many percentage points and by how many percent has this share changed?

Solution (1/2) – Frequency Table

Rule

h_j = number of values equal to x_j; $f_j = h_j/n$; $F(x_j)$ = sum of all f_l with $x_l \leq x_j$.

a) Count the $n = 12$ values (check: the h_j add up to 12, the f_j to 1):

Stars x_j	2	3	4	5	Total
Absolute frequency h_j	1	2	4	5	12
Relative frequency f_j (fraction)	$\frac{1}{12}$	$\frac{2}{12}$	$\frac{4}{12}$	$\frac{5}{12}$	1
Relative frequency f_j (decimal)	0.0833	0.1667	0.3333	0.4167	1
Cumulative rel. frequency $F(x_j)$	0.0833	**0.2500**	0.5833	1	

b) "At most 3 stars" = cumulative relative frequency at $x = 3$:
$F(3) = f_1 + f_2 = \frac{1}{12} + \frac{2}{12} = \frac{3}{12} = \mathbf{0.25}$ $(x_1 = 2, x_2 = 3) \rightarrow$ a quarter of the ratings.

Solution (2/2) – Frequency Table

c) "At least 4 stars" = complement of "at most 3": $1 - F(3) = 1 - 0.25 = \mathbf{0.75}$, i.e. $h_3 + h_4 = 4 + 5 = \mathbf{9}$ ratings.
d) Largest absolute frequency $h_4 = 5$ at $x_4 = 5 \rightarrow$ the most frequent rating is **5** stars.
e) This year $f_4 = \frac{5}{12} = 41.67\,\%$ (last year: $30\,\%$):

- absolute difference: $41.67\,\% - 30\,\% = \mathbf{11.67}$ **percentage points**
- relative change: $(41.67\,\% - 30\,\%)/30\,\% = 0.389 = \mathbf{38.9}\,\%$

```
stars = pd.Series([4, 5, 3, 4, 5, 5, 2, 4, 5, 3, 4, 5])
print(stars.value_counts().sort_index().tolist())      # [1, 2, 4, 5]
F = stars.value_counts(normalize=True).sort_index().cumsum()
print(F.round(4).tolist())                             # [0.0833, 0.25, 0.5833, 1.0]
print((stars <= 3).mean(), (stars >= 4).sum())         # 0.25 9
```

Interpretation

Three quarters of the guests give 4 or 5 stars; the share of top ratings has grown.

▶ **Try it yourself:** notebook, section 2.2 "Exercises" – open in Colab

Measures of Central Tendency

Measures of Central Tendency – Means, Median and Quantiles

Measures of Central Tendency

Guiding question

Which shoe size is "typical" for the soccer team – and how do we summarise eleven values in a single number?

- Where is the middle?
- Where is the centre of gravity of the distribution?
- At which value is one quarter of the values reached?
- At which value are three quarters of the values reached?

Refresher: Summation and Product Signs

Example data (first three players): $x_1 = 40$, $x_2 = 39$, $x_3 = 38$

Summation sign Σ

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \cdots + x_n$$

Read: "sum of the x_i for i from 1 to n"

Example: $\sum_{i=1}^{3} x_i = 40 + 39 + 38 = 117$

Product sign Π

$$\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdots x_n$$

Read: "product of the x_i for i from 1 to n"

Example: $\prod_{i=1}^{3} x_i = 40 \cdot 39 \cdot 38 = 59\,280$

Common mistake

$\sum x_i^2 \neq \left(\sum x_i\right)^2$, e.g. $1^2 + 2^2 = 5 \neq (1+2)^2 = 9$: first square, then add (as in the variance later)!

Means (1): Arithmetic Mean and Trimmed Mean

Arithmetic mean

Sum of all values divided by n: $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

In words: the value every unit would have if the total were shared out equally.

Advantage: easy to compute. **Disadvantage:** sensitive to extreme values (outliers).

Trimmed mean $\bar{x}_{\text{trim}}$

Choose a share α (e.g. 10 %) to cut off at each end: remove the $\lfloor \alpha \cdot n \rfloor$ smallest and the $\lfloor \alpha \cdot n \rfloor$ largest values ($\lfloor \cdot \rfloor$ = round down), then take the arithmetic mean of the rest.

Example (shoe sizes, $\alpha = 10\,\%$): $\lfloor 0.1 \cdot 11 \rfloor = 1$ value per end $\rightarrow$ remove 37 and 46:

$$\bar{x}_{\text{trim}} = \frac{436 - 37 - 46}{9} = \frac{353}{9} = 39.2222$$

Means (2): Geometric Mean – Averaging Growth Rates

Geometric mean (only for positive values)

Product of all values, n-th root: $\bar{x}_{\text{geo}} = \left(\prod_{i=1}^{n} x_i\right)^{1/n} = \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$

Motivation: revenue grows by $r_1 = +20\,\%$ in year 1 and by $r_2 = -10\,\%$ in year 2. Is the average growth $(20\,\% - 10\,\%)/2 = 5\,\%$ per year? **No!**

- Growth rate $r_i \rightarrow$ **growth factor** $1 + r_i$: 1.20 and 0.90
- Factors **multiply**: $1.20 \cdot 0.90 = 1.08$, i.e. $+8\,\%$ in total
- Average factor: $\bar{x}_{\text{geo}} = \sqrt{1.08} = 1.0392 \rightarrow +\mathbf{3.92}\,\%$ per year
- Check: $1.0392^2 \approx 1.08$ ✓, whereas $1.05^2 = 1.1025 \neq 1.08$

Key takeaway

Average growth rate $= \bar{x}_{\text{geo}}$ of the growth factors $1 + r_i$, minus 1. $\bar{x}_{\text{geo}}$ is the *constant* factor that produces the same total growth.

Median

Definition

At or below which observation do at least 50 % of the values lie?

Computation: sort the data; the median is the value at position $\lceil n/2 \rceil$: $\tilde{x} = x_{(\lceil n/2 \rceil)}$

Note: alternative definition

In some textbooks, the median for even n is defined as the average of the two middle values (the "middle value"). In this course, we use the $\lceil n/2 \rceil$ definition.

Advantages of the $\lceil n/2 \rceil$ definition:

- The median is always an actual observation
- Consistent with the quantile definition: Median = 50 %-quantile $= x_{(\lceil 0.5 \cdot n \rceil)}$

Advantages of the median in general:

- Robust to outliers
- Also suitable for ordinal data

Mode

Definition

The most frequent value in the distribution

- **Advantage:** suitable for both ordinal and nominal data
- **Disadvantage:** often not unique
- **Disadvantage:** may not represent the "centre" of the distribution

Which Measure for Which Level of Measurement?

Level	Mode	Median, quantiles	Arithmetic mean
Nominal	✓	–	–
Ordinal	✓	✓	–
Metric	✓	✓	✓

- Median and quantiles need a **ranking**, the mean needs meaningful **differences**.
- The geometric mean needs positive values on a ratio scale (e.g. growth factors).

Back to the online retailer (exercise in the Fundamentals)

Mean (and median, mode): number of items, order value, delivery time, order date
Median (and mode), no mean: customer rating, customer status
Only the mode: postcode, payment method

Measures of Central Tendency – Example

Player	1	2	3	4	5	6	7	8	9	10	11
Shoe size	40	39	38	41	46	39	39	38	37	40	39

Measure	**Value**
Sum	436
Arithmetic mean	39.6364
Geometric mean	39.5755
Trimmed mean (10 % per end)	39.2222
Median	39
Mode	39

Exercise – Mean, Median and Mode

Problem statement

An electronics shop records the number of laptops sold on $n = 9$ days:

$$4,\ 7,\ 3,\ 5,\ 4,\ 6,\ 4,\ 2,\ 10$$

a) Compute the arithmetic mean, the median, the mode and the trimmed mean with $\alpha = 20\,\%$.

b) On day 9 the shop actually sold 19 laptops (a large company order), not 10. Which of the four measures change, and to which value?

c) $n = 7$ customers rate the service: good, very good, fair, good, poor, good, very good. Which of the measures may be used for this variable? Determine them.

Solution (1/2) – Mean, Median and Mode

Rule

Sort first; $\bar{x} = \frac{1}{n}\sum x_i$; median $= x_{(\lceil n/2 \rceil)}$; mode = most frequent value;
trimmed mean: drop $\lfloor \alpha \cdot n \rfloor$ values at each end, then average the rest.

a) Sorted data: 2, 3, 4, 4, 4, 5, 6, 7, 10 (sum $= 45$)

Calculation

$$\bar{x} = \frac{45}{9} = \mathbf{5}, \qquad \tilde{x} = x_{(\lceil 9/2 \rceil)} = x_{(5)} = \mathbf{4}, \qquad \text{mode} = \mathbf{4} \text{ (occurs 3 times)}$$

$$\bar{x}_{\text{trim}} = \frac{45 - 2 - 10}{7} = \frac{33}{7} = \mathbf{4.7143} \qquad (\lfloor 0.2 \cdot 9 \rfloor = \lfloor 1.8 \rfloor = 1 \text{ value removed per end})$$

b) With 19 instead of 10 the sum grows by 9 to 54 $\rightarrow \bar{x} = 54/9 = \mathbf{6}$ (up by 1). The order is unchanged (19 is still the largest value) $\rightarrow$ median $= \mathbf{4}$, mode $= \mathbf{4}$ and $\bar{x}_{\text{trim}} = (54 - 2 - 19)/7 = \mathbf{4.7143}$ stay the same.

Solution (2/2) – Mean, Median and Mode

c) The service rating is **ordinal**: the categories have a ranking (poor $<$ fair $<$ good $<$ very good), but the distances between them are not meaningful.

- Sorted: poor, fair, good, good, good, very good, very good
- Median: $x_{(\lceil 7/2 \rceil)} = x_{(4)} =$ **good**
- Mode: **good** (3 of the 7 ratings)
- Arithmetic (and trimmed) mean: **not meaningful** – "(good + poor) / 2" is not defined, and the step from poor to fair need not equal the step from good to very good.

Interpretation

A single extreme day moves the mean ($5 \to 6$), but not the median, the mode or the trimmed mean: these are **robust** to outliers. For ordinal data, use the median and the mode.

Arithmetic Mean from a Frequency Table

Idea: if the data are given as a frequency table, each value x_j enters the sum h_j times.

Rule

$$\bar{x} = \frac{1}{n}\sum_{j=1}^{k} h_j \cdot x_j = \sum_{j=1}^{k} f_j \cdot x_j \qquad (k = \text{number of distinct values})$$

Shoe size x_j	37	38	39	40	41	46	Total
h_j	1	2	4	2	1	1	11
$h_j \cdot x_j$	37	76	156	80	41	46	436

$\bar{x} = \frac{436}{11} = \mathbf{39.6364}$ – the same result as from the raw data.

In words: a *weighted* mean: every value counts as often as it occurs; the weights are the relative frequencies f_j.

Measures of Central Tendency with Python (1)

```python
import numpy as np
from scipy import stats
# Sum
print(data.sum())                    # 436
# Arithmetic mean
print(data.mean())                   # 39.6364
# Geometric mean
# Step 1: Calculate the product of all values
product = np.prod(data)
# Step 2: Determine the number of observations
n = len(data)
# Step 3: Take the n-th root
gmean = product ** (1 / n)           # 39.5755
print(stats.gmean(data))             # same result in one step
```

- **Try it yourself:** notebook, section 2.3 "Measures of Central Tendency" – open in Colab

Measures of Central Tendency with Python (2)

```
# Trimmed mean: 10% of the values are removed at EACH end
print(stats.trim_mean(data, 0.10))  # 39.2222

# Median
print(np.median(data))              # 39.0

# Mode (all most frequent values, as a list)
print(data.mode().tolist())         # [39]

# Mean from a frequency table: weighted mean
print(np.average([37, 38, 39, 40, 41, 46], weights=[1, 2, 4, 2, 1, 1]))  # 39.6364
```

Note: NumPy's median definition

For even n, `np.median()` averages the two middle values – the course definition $x_{(\lceil n/2 \rceil)}$ is obtained with `np.quantile(data, 0.5, method="inverted_cdf")`.

▶ **Try it yourself:** notebook, section 2.3 "Measures of Central Tendency" – open in Colab

Quantiles

First, the values must be sorted!

1st quartile: At or below which value do at least 25 % of the values lie?

2nd quartile: = Median (at or below which value do at least 50 % of the values lie?)

3rd quartile: At or below which value do at least 75 % of the values lie?

Arbitrary p-quantile: At or below which value do at least $p \cdot 100$ % of the values lie?

Quantiles – Computation

Rule (course definition)

$$x_p = x_{(\lceil n \cdot p \rceil)}$$

Compute $n \cdot p$ and round **up** to the next integer: this is the position in the sorted data. If $n \cdot p$ is already an integer, it stays as it is.

In words: count $\lceil n \cdot p \rceil$ values from the bottom of the sorted list – at least $p \cdot 100\,\%$ of the observations lie at or below this value.

Example 80 %-quantile ($n = 11$): $n \cdot 0.8 = 11 \cdot 0.8 = 8.8$, round up to position 9 $\rightarrow$ $x_{0.8} = x_{(9)} = 40$.

Example $n = 10$, $p = 0.2$: $n \cdot p = 2$ is already an integer $\rightarrow$ position 2.

Quantiles – Computation Example

Quantile	p	$n \cdot p$	Position	Value
1st quartile	0.25	2.75	3	38
2nd quartile	0.50	5.50	6	39
3rd quartile	0.75	8.25	9	40
17 %-quantile	0.17	1.87	2	38
80 %-quantile	0.80	8.80	9	40
5 %-quantile	0.05	0.55	1	37

Quantiles – Worked Example

Sorted data: 37, 38, 38, 39, 39, 39, 39, 40, 40, 41, 46

$$\text{Median} = x_{(\lceil 0.5 \cdot 11 \rceil)} = x_{(6)} = 39$$
$$Q_1 = x_{(\lceil 0.25 \cdot 11 \rceil)} = x_{(3)} = 38$$
$$Q_3 = x_{(\lceil 0.75 \cdot 11 \rceil)} = x_{(9)} = 40$$

Exercise – Quantiles of Hotline Waiting Times

Problem statement

A customer hotline measures the waiting time (in minutes, rounded) of $n = 12$ calls:

$$3,\ 1,\ 5,\ 2,\ 6,\ 4,\ 2,\ 8,\ 3,\ 4,\ 2,\ 5$$

a) Sort the data.

b) Determine the 10 %-quantile, Q_1, the median, Q_3 and the 90 %-quantile.

c) The hotline wants to promise: "At least 90 % of our callers wait at most x minutes." What is the smallest x supported by the data?

d) A colleague reports a median of 3.5 minutes. How did they obtain this value, and what is the median according to the course definition?

Solution (1/2) – Quantiles of Hotline Waiting Times

Rule

$x_p = x_{(\lceil n \cdot p \rceil)}$: round $n \cdot p$ **up** (an integer stays as it is); take the value at this position.

a) Sorted data $(x_{(1)}, \ldots, x_{(12)})$: 1, 2, 2, 2, 3, 3, 4, 4, 5, 5, 6, 8

b) Quantiles for $n = 12$:

Quantile	p	$n \cdot p$	Position	Value
10 %-quantile	0.10	1.2	2	$x_{(2)} = \mathbf{2}$
Q_1	0.25	3.0	3	$x_{(3)} = \mathbf{2}$
Median	0.50	6.0	6	$x_{(6)} = \mathbf{3}$
Q_3	0.75	9.0	9	$x_{(9)} = \mathbf{5}$
90 %-quantile	0.90	10.8	11	$x_{(11)} = \mathbf{6}$

Note: $x_{0.1} = Q_1 = 2$ because of ties (three calls of exactly 2 minutes) – only the position counts.

Solution (2/2) – Quantiles of Hotline Waiting Times

c) The promise asks for the 90 %-quantile: $x_{0.9} = x_{(11)} = \mathbf{6}$ minutes.

- Check with the cumulative relative frequency: $F(6) = \frac{11}{12} = 0.9167 \geq 0.9$ ✓
- 5 minutes are not enough: $F(5) = \frac{10}{12} = 0.8333 < 0.9$

d) The colleague averaged the two middle values $x_{(6)} = 3$ and $x_{(7)} = 4$: $(3 + 4)/2 = 3.5$ – the alternative textbook definition for even n. The course definition gives $\tilde{x} = x_{(\lceil 12/2 \rceil)} = x_{(6)} = \mathbf{3}$ minutes.

Interpretation

At least half of the callers wait at most 3 minutes and at least 90 % at most 6 minutes – only the single caller with 8 minutes waits longer.

Quantiles with Python

```python
# Course definition x_(ceil(n*p)): method="inverted_cdf"
# All standard quantiles at once (0, 25, 50, 75, 100 %)
p = [0, 0.25, 0.50, 0.75, 1]
print(np.quantile(data, p, method="inverted_cdf"))   # [37 38 39 40 46]

# 1st quartile (25%-quantile)
print(np.quantile(data, 0.25, method="inverted_cdf"))  # 38

# 80%-quantile
print(np.quantile(data, 0.80, method="inverted_cdf"))  # 40
```

Pitfall: the default method interpolates

`np.quantile(data, 0.25)` and `data.quantile(0.25)` use linear interpolation and return 38.5 – always pass `method="inverted_cdf"` to get the course definition.

▶ **Try it yourself:** notebook, section 2.3 "Measures of Central Tendency" – open in Colab

Exercise – Measures of Central Tendency and Quantiles

Problem statement

Annual salaries (in thousand euros) of the $n = 10$ employees of a department, including the head of department:

$$34,\ 42,\ 28,\ 120,\ 36,\ 31,\ 45,\ 34,\ 50,\ 39$$

a) Sort the data and compute the arithmetic mean, the median and the mode.

b) Compute the trimmed mean (remove 10 % of the values at each end).

c) Determine Q_1, Q_3 and the 20 %- and 90 %-quantiles.

d) Which measure best describes the "typical" salary in the department? Give reasons.

Solution (1/4) – Measures of Central Tendency and Quantiles

Rule

Sort first; $\bar{x} = \frac{1}{n}\sum x_i$; median $= x_{(\lceil n/2 \rceil)}$; mode = most frequent value.

a) Sorted data $(x_{(1)}, \ldots, x_{(10)})$: 28, 31, 34, 34, 36, 39, 42, 45, 50, 120

Calculation

$$\bar{x} = \frac{28 + 31 + 34 + 34 + 36 + 39 + 42 + 45 + 50 + 120}{10} = \frac{459}{10} = \mathbf{45.9}$$

$$\tilde{x} = x_{(\lceil 0.5 \cdot 10 \rceil)} = x_{(5)} = \mathbf{36}$$

Mode = **34** (the only value that occurs twice)

The arithmetic mean (45.9) lies far above the median (36): 8 of the 10 employees earn less than the average.

Solution (2/4) – Measures of Central Tendency and Quantiles

b) Trimmed mean: $\lfloor 0.1 \cdot 10 \rfloor = 1$ value per end $\rightarrow$ remove 28 and 120, average the rest:

$$\bar{x}_{\text{trim}} = \frac{31 + 34 + 34 + 36 + 39 + 42 + 45 + 50}{8} = \frac{311}{8} = \mathbf{38.875}$$

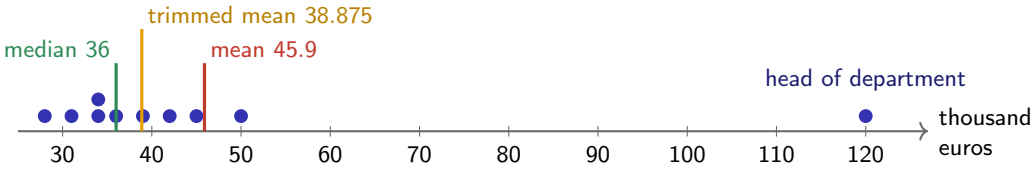
c) Rule: position $= \lceil n \cdot p \rceil$ in the sorted data; if $n \cdot p$ is already an integer, it is not rounded up.

Quantile	p	$n \cdot p$	Position	Value
20 %-quantile	0.20	2.0	2	$x_{(2)} = \mathbf{31}$
Q_1	0.25	2.5	3	$x_{(3)} = \mathbf{34}$
Q_3	0.75	7.5	8	$x_{(8)} = \mathbf{45}$
90 %-quantile	0.90	9.0	9	$x_{(9)} = \mathbf{50}$

Note: $x_{(3)} = x_{(4)} = 34$ – ties (equal values) do not change the rule; only the position counts.

Solution (3/4) – Measures of Central Tendency and Quantiles

d) The salary of the head of department (120) is an outlier and pulls the mean up to 45.9 – a salary hardly anyone in the department earns. The **median (36)** and the trimmed mean (38.9) are robust to the outlier and describe the typical salary better.



Key takeaway

For skewed data with outliers (salaries, wealth, sales) always report the median **and** the mean – the gap between them reveals the skewness.

Solution (4/4) – Measures of Central Tendency and Quantiles

```
# Python check; course quantiles need method="inverted_cdf"
salary = np.array([34, 42, 28, 120, 36, 31, 45, 34, 50, 39])
print(np.sort(salary))
print(salary.mean(), stats.trim_mean(salary, 0.10))
print(pd.Series(salary).mode().tolist())
print(np.quantile(salary, 0.5, method="inverted_cdf"), np.median(salary))
print(np.quantile(salary, [0.2, 0.25, 0.75, 0.9], method="inverted_cdf"))
```

```
[ 28  31  34  34  36  39  42  45  50 120]
45.9 38.875
[34]
36 37.5
[31 34 45 50]
```

Pitfall: even n

`np.median()` returns $37.5 = (x_{(5)} + x_{(6)})/2$; the course median is $x_{(\lceil 10/2 \rceil)} = x_{(5)} = 36$.

▶ **Try it yourself:** notebook, section 2.4 "Exercises" – open in Colab

Exercise – Geometric Mean

Problem statement

The revenue of a start-up was exactly 200 thousand euros in year 0. In the following three years it changed by

$$+10\,\%, \qquad -5\,\%, \qquad +20\,\%.$$

a) What is the revenue after three years?

b) Compute the arithmetic mean of the three growth rates. Which revenue after three years would you obtain if revenue had grown by this mean every year?

c) Compute the average annual growth rate via the geometric mean of the growth factors and check the result.

d) Why does the arithmetic mean in b) give a wrong result?

Solution (1/2) – Geometric Mean

Rule

Growth rate $r_i \rightarrow$ growth factor $1 + r_i$. Several periods: **multiply** the factors.

a) Growth factors: 1.10; 0.95; 1.20

$$\text{Revenue}_3 = 200 \cdot 1.10 \cdot 0.95 \cdot 1.20 = 200 \cdot 1.254 = \mathbf{250.8} \text{ thousand euros}$$

b) Arithmetic mean of the rates:

$$\bar{r} = \frac{0.10 - 0.05 + 0.20}{3} = \frac{0.25}{3} = 0.0833 = \mathbf{8.33}\,\%$$

With this mean: $200 \cdot 1.0833^3 = 200 \cdot 1.2714 = \mathbf{254.3}$ thousand euros $\neq 250.8 \rightarrow$ **too high!**

Solution (2/2) – Geometric Mean

c) Geometric mean of the growth factors:

$$\bar{x}_{\text{geo}} = (1.10 \cdot 0.95 \cdot 1.20)^{1/3} = 1.254^{1/3} = \mathbf{1.0784} \quad \Rightarrow \quad \bar{r}_{\text{geo}} = 1.0784 - 1 = \mathbf{7.84}\,\%$$

Check: $200 \cdot 1.0784^3 \approx 200 \cdot 1.254 = 250.8$ thousand euros ✓

d) Growth acts **multiplicatively**: the $+20\,\%$ applies to a different base than the $+10\,\%$. The arithmetic mean of the rates is never smaller than the geometric one – it overstates growth.

```
factors = np.array([1.10, 0.95, 1.20])
print(200 * np.prod(factors))        # 250.8
print(factors.mean() - 1)            # 0.0833  (arithmetic: wrong)
print(stats.gmean(factors) - 1)      # 0.0784  (geometric: correct)
```

Key takeaway

Average growth rates, returns, price increases: **geometric mean of the growth factors**.

▶ **Try it yourself:** notebook, section 2.4 "Exercises" – open in Colab

Measures of Dispersion

Measures of Dispersion – Range, Interquartile Range,
Variance, Standard Deviation,
Coefficient of Variation, Normalised Coefficient of Variation

Measures of Dispersion – Why?

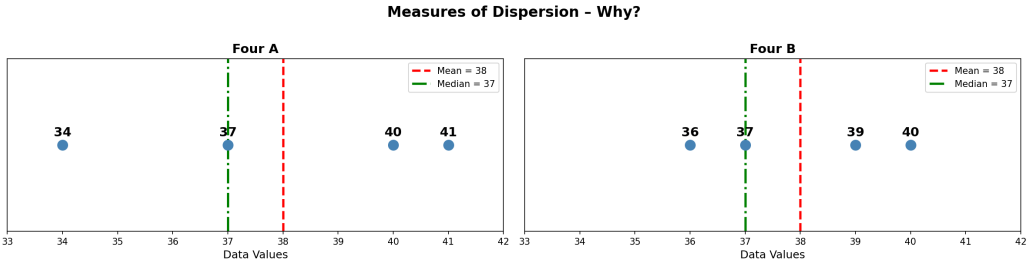
Example: Shoe sizes of two women's rowing fours (without coxswain)

	Four A	**Four B**
Data	34, 37, 40, 41	36, 37, 39, 40
Mean	38	38
Median	37	37

Question

The mean and median are identical. Does that mean the data sets are equivalent?

Measures of Dispersion – Visualisation



- Mean and median are identical – but the **dispersion** differs considerably
- Four A: Data farther from the mean $\rightarrow$ greater dispersion
- Four B: Data closely concentrated around the mean $\rightarrow$ less dispersion

Range and Interquartile Range

Definitions

Range: $R = x_{(n)} - x_{(1)}$ (largest minus smallest value)

Interquartile range: $\text{IQR} = Q_3 - Q_1 = x_{0.75} - x_{0.25}$

In words: R is the width of the whole data set, the IQR the width of its middle 50 %.

Example (soccer team, sorted): 37, 38, **38**, 39, 39, 39, 39, 40, **40**, 41, 46

- $R = 46 - 37 = \mathbf{9}$, $\text{IQR} = x_{(9)} - x_{(3)} = 40 - 38 = \mathbf{2}$
- Rowing fours: $R_A = 41 - 34 = 7 > R_B = 40 - 36 = 4$ and $\text{IQR}_A = 40 - 34 = 6 > \text{IQR}_B = 39 - 36 = 3$

Range vs. IQR

R depends only on the two most extreme values: the player with size 46 alone makes $R = 9$. The IQR ignores the outer 25 % on each side and is **robust** to outliers.

Sample Variance

Definition

Mean squared deviation from the mean (corrected with $n-1$):

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad 0 \leq s^2 < \infty$$

In words: roughly the average squared distance of the values from their mean.

- **Why square?** – The deviations from the mean always cancel out exactly, $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$. Squaring makes every deviation count positively.
- **Why divide by $n-1$ only?** – Because using the mean leaves only $n-1$ degrees of freedom: if the mean and $n-1$ values are known, the n-th value can be computed.
- The dimension has changed, e.g. cm becomes cm^2.
- Comparison requires the same units. With different n, the precision of the estimates may differ.

Sample Variance – Worked Example

	Four A	Four B
Data	34, 37, 40, 41	36, 37, 39, 40
Mean $\bar{x}$	38	38
Deviations $x_i - \bar{x}$	$-4,\ -1,\ +2,\ +3$	$-2,\ -1,\ +1,\ +2$

Check: the deviations sum to 0 in both cases.

Calculation

$$s_A^2 = \frac{1}{4-1}\left[(34-38)^2 + (37-38)^2 + (40-38)^2 + (41-38)^2\right] = \frac{30}{3} = 10$$

$$s_B^2 = \frac{1}{4-1}\left[(36-38)^2 + (37-38)^2 + (39-38)^2 + (40-38)^2\right] = \frac{10}{3} \approx 3.33$$

Standard Deviation

Definition

The standard deviation is the square root of the variance:

$$s = +\sqrt{s^2} \qquad 0 \leq s < \infty$$

- By taking the square root, the measure of dispersion is again in the **dimension of the original values**
- If $s = 0$: no dispersion – all values are equal and equal the mean
- Same unit as the original values, therefore **more intuitive** than the sample variance
- Comparison requires the same units. With different n, the precision of the estimates may differ.

Standard Deviation – Worked Example

Calculation

$$s_A = +\sqrt{s_A^2} = +\sqrt{10} = 3.1623$$

$$s_B = +\sqrt{s_B^2} = +\sqrt{3.\overline{3}} = 1.8257$$

Interpretation

The values of Four A scatter around the mean with a standard deviation of about 3.16 shoe sizes, the values of Four B only with about 1.83.
$\rightarrow$ The dispersion is now expressed in the **same unit** as the original data.

Exercise – Variance and Standard Deviation

Problem statement

The online retailer compares two parcel services. Delivery times (in days) of $n = 5$ orders each:

Order i	1	2	3	4	5
Parcel service A	2	5	1	3	4
Parcel service B	4	3	5	4	4

a) Compute the mean $\bar{x}$ and the range R for both services.

b) Compute the sample variance s^2 and the standard deviation s (with units).

c) Which service delivers faster on average, which more reliably? Give reasons.

Solution (1/2) – Variance and Standard Deviation

Rule

Deviations from the mean, square them, add them up, divide by $n-1=4$:
$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2$; $\quad s = +\sqrt{s^2}$.

a) Means: $\bar{x}_A = \frac{2+5+1+3+4}{5} = \frac{15}{5} = \mathbf{3}$ days, $\quad \bar{x}_B = \frac{4+3+5+4+4}{5} = \frac{20}{5} = \mathbf{4}$ days

Ranges: $R_A = 5-1 = \mathbf{4}$ days, $\quad R_B = 5-3 = \mathbf{2}$ days

Order i	1	2	3	4	5	Total
A: $x_i - 3$	-1	2	-2	0	1	0
A: $(x_i-3)^2$	1	4	4	0	1	10
B: $x_i - 4$	0	-1	1	0	0	0
B: $(x_i-4)^2$	0	1	1	0	0	2

Check: the deviations sum to 0 for both services ✓

Solution (2/2) – Variance and Standard Deviation

b) Divide the sums of squares by $n - 1 = 4$, then take the square root:

$$s_A^2 = \frac{10}{4} = \mathbf{2.5}\ \text{days}^2, \qquad s_A = \sqrt{2.5} = \mathbf{1.5811}\ \text{days}$$

$$s_B^2 = \frac{2}{4} = \mathbf{0.5}\ \text{days}^2, \qquad s_B = \sqrt{0.5} = \mathbf{0.7071}\ \text{days}$$

c) Comparison:

- **Faster on average:** service **A** ($\bar{x}_A = 3$ days vs. $\bar{x}_B = 4$ days)
- **More reliable:** service **B** ($s_B = 0.71$ vs. $s_A = 1.58$ days; $R_B = 2$ vs. $R_A = 4$ days): its delivery times stay close to the mean, those of A range from 1 to 5 days

Interpretation

To promise a fixed delivery date, choose B; to be fast on average, choose A. The standard deviation can be read directly in **days**, the variance (in days^2) cannot.

Variance from a Frequency Table

Rule: as for the mean, every squared deviation enters the sum h_j times

$$s^2 = \frac{1}{n-1} \sum_{j=1}^{k} h_j \cdot (x_j - \bar{x})^2$$

Soccer team: x_j	37	38	39	40	41	46	Total
h_j	1	2	4	2	1	1	11
$x_j - \bar{x}$ $(\bar{x} = 39.6364)$	-2.6364	-1.6364	-0.6364	0.3636	1.3636	6.3636	
$h_j \cdot (x_j - \bar{x})^2$	6.9504	5.3554	1.6198	0.2645	1.8595	40.4959	56.5455

$$s^2 = 56.5455/10 = \mathbf{5.6545}, \qquad s = \sqrt{5.6545} = \mathbf{2.3779}$$

Observation: the single value 46 contributes 40.50 of 56.55 (72 %) to the sum of squares $\rightarrow$ s^2 and s react strongly to outliers.

Coefficient of Variation

Definition

The coefficient of variation is a **relative** measure of dispersion:

$$v = \frac{s}{\bar{x}} \qquad \text{for data with } x_i > 0 \text{ for all } i \text{ (hence } \bar{x} > 0)$$

- **Relative** means: independent of the scale (order of magnitude)
- The values must all be positive (ratio scale with a natural zero)
  - Mathematically: $\bar{x} > 0$, since otherwise division by zero or a negative v
  - Substantively: $\bar{x}$ must reflect a meaningful **level** of the data
  - With mixed signs, $\bar{x} \approx 0$ is possible – then v becomes extremely large, even though the dispersion is not unusual
  - Example: $-10, -5, +5, +11 \Rightarrow \bar{x} = 0.25$, v loses its interpretive value
- Typical applications: incomes, weights, prices
- However: with very different sample sizes, the precision of the estimates differs, so comparisons should be interpreted with caution

Coefficient of Variation – Worked Example

Calculation

$$v_A = \frac{s_A}{\bar{x}_A} = \frac{3.1623}{38} = 0.0832$$

$$v_B = \frac{s_B}{\bar{x}_B} = \frac{1.8257}{38} = 0.0480$$

Interpretation

The standard deviation of Four A amounts to about 8.3 % of its mean, that of Four B to only about 4.8 %.

$\rightarrow$ As a relative measure, v allows comparison of dispersions **regardless of the scale** of the data.

Normalised Coefficient of Variation

Definition

The normalised coefficient of variation is a relative measure of dispersion **normalised** to the interval $[0, 1]$:

$$v^* = \frac{s}{\bar{x}} \cdot \frac{1}{\sqrt{n}} = \frac{v}{\sqrt{n}} \qquad 0 \leq v^* \leq 1$$

- Same prerequisites as for the coefficient of variation v
- **Background:** for non-negative data, v is at most $\sqrt{n}$ (extreme case: all values but one are 0). Dividing by $\sqrt{n}$ therefore puts v on a scale from 0 to 1.
- **Advantage over** v: $v^* = 0$ means no dispersion, $v^* = 1$ the maximum possible dispersion $\rightarrow$ one can judge whether the relative dispersion is small or large **in absolute terms**
- **Caution:** v^* is **not** suitable for comparing samples of different size: with the same dispersion, v^* gets smaller just because n is larger ($\propto 1/\sqrt{n}$)

Normalised Coefficient of Variation – Worked Example

Calculation

$$v_A^* = \frac{v_A}{\sqrt{n}} = \frac{0.0832}{\sqrt{4}} = \frac{0.0832}{2} = 0.0416$$

$$v_B^* = \frac{v_B}{\sqrt{n}} = \frac{0.0480}{\sqrt{4}} = \frac{0.0480}{2} = 0.0240$$

Interpretation

On the scale from 0 to 1 both values are close to 0 $\rightarrow$ the relative dispersion is very low in both fours; the sizes in Four A vary relatively more than in Four B.

$\rightarrow$ Since both fours have the same $n = 4$, normalising does not change the comparison: $v_A^*/v_B^* = v_A/v_B$. With different n, a comparison via v^* would not be meaningful.

Overview of Measures of Dispersion

	Four A	Four B
Mean	38	38
Median	37	37
Range	7	4
Interquartile range	6	3
Sample variance	10	3.3333
Standard deviation	3.1623	1.8257
Coefficient of variation	0.0832	0.0480
Normalised coefficient of variation	0.0416	0.0240

Measures of Dispersion with Python

```
four_a = np.array([34, 37, 40, 41])
# Range and interquartile range (course quantiles)
print(np.ptp(four_a))                                          # 7
q1, q3 = np.quantile(four_a, [0.25, 0.75], method="inverted_cdf")
print(q3 - q1)                                                 # 6
# Sample variance (ddof=1: divide by n-1) and standard deviation
print(np.var(four_a, ddof=1), np.std(four_a, ddof=1))    # 10.0 3.1623
# Coefficient of variation and normalised coefficient of variation
v = np.std(four_a, ddof=1) / np.mean(four_a)                   # 0.0832
vstar = v / np.sqrt(len(four_a))                               # 0.0416
```

Pitfall: NumPy divides by n by default

`np.var(four_a)` uses `ddof=0` and returns 7.5 instead of $s^2 = 10$ – always pass `ddof=1`. pandas' `.var()` and `.std()` already use `ddof=1`.

▶ **Try it yourself:** notebook, section 2.5 "Measures of Dispersion" – open in Colab

Exercise – Which Measure of Dispersion for Which Comparison?

Problem statement

Which measures of dispersion (s^2, s, v, v^*) are suitable for comparing

1. the shoe sizes of the rowing four and the rowing eight?
2. the shoe sizes of the women's four and the men's four?
3. the shoe sizes of the women's four and the women's four with coxswain?
4. the shoe sizes of the women's four and the men's eight?

Hint – Which Measure of Dispersion?

When is which measure suitable?

The sample size affects s^2, s and v only through the **precision of the estimate**, not the measure itself – an eight with the same dispersion as a four has (apart from random fluctuations) the same s and the same v. Only v^* gets smaller just because n is larger.

Measure of dispersion	**Scale (order of magnitude)**	**Sample size n**
Sample variance s^2	dependent	measure independent of n
Standard deviation s	dependent	measure independent of n
Coefficient of variation v	independent	measure independent of n
Norm. coeff. of variation v^*	independent	decreases with n ($\propto 1/\sqrt{n}$)

Solution (1/3) – Which Measure of Dispersion?

1. **Four vs. eight** (same unit, same scale, **different** n)
   - s^2, s, v: **yes** – the measures do not depend on n; the different n (4 vs. 8) only affects the precision of the estimate
   - v^*: **no** – the eight would get a smaller v^* just because $\sqrt{8} > \sqrt{4}$, even with the same dispersion
2. **Women's four vs. men's four** (same unit, **different scale**, same n)
   - s^2, s: less suitable, since men's shoe sizes are systematically larger
   - v: **yes** – not affected by the scale
   - v^*: **yes** – with equal n only a rescaling of v

Solution (2/3) – Which Measure of Dispersion?

3. **Women's four vs. women's four with coxswain** (same unit, same scale, **different** n)
   - s^2, s, v: **yes** – measures independent of n; the different n (4 vs. 5) only affects the precision of the estimate
   - v^*: **no** – divided by $\sqrt{5}$ instead of $\sqrt{4}$, the comparison is distorted
4. **Women's four vs. men's eight** (**different scale** and **different** n)
   - s^2, s: less suitable, the scale differs
   - v: **yes** – not affected by the scale and independent of n
   - v^*: **no** – the different n distorts the comparison

Solution (3/3) – Which Measure of Dispersion?

1. The shoe sizes of the rowing four and the rowing eight.
2. The shoe sizes of the women's four and the men's four.
3. The shoe sizes of the women's four and the women's four with coxswain.
4. The shoe sizes of the women's four and the men's eight.

	1.	**2.**	**3.**	**4.**
Sample variance	yes	no	yes	no
Standard deviation	yes	no	yes	no
Coefficient of variation	yes	yes	yes	yes
Normalised coefficient of variation	no	yes	no	no

Exercise – Measures of Dispersion of the Handball Team

Problem statement

Shoe sizes of the handball team ($n = 13$):
40, 39, 38, 41, 46, 39, 39, 38, 37, 40, 39, 40, 42

Calculate the following measures:

Measure	Formula	Value
Mean $\bar{x}$	$\frac{1}{n}\sum_{i=1}^{n} x_i$	?
Sample variance s^2	$\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	?
Standard deviation s	$+\sqrt{s^2}$	?
Coefficient of variation v	$s/\bar{x}$	?
Norm. coeff. of variation v^*	$v/\sqrt{n}$	?

Solution – Measures of Dispersion of the Handball Team

Intermediate values: $\sum x_i = 518, \quad \sum (x_i - \bar{x})^2 = 61.6923$

Measure	Calculation	Value
Mean $\bar{x}$	$518/13$	**39.8462**
Sample variance s^2	$61.6923/12$	**5.1410**
Standard deviation s	$\sqrt{5.1410}$	**2.2674**
Coefficient of variation v	$2.2674/39.8462$	**0.0569**
Norm. coeff. of variation v^*	$0.0569/\sqrt{13}$	**0.0158**

Interpretation

Soccer team: $s = 2.3779$, $v = 2.3779/39.6364 = 0.0600 \rightarrow$ the shoe sizes of both teams vary similarly. Their v^* (0.0158 vs. 0.0181) must **not** be compared: n differs (13 vs. 11).

Exercise – Comparing the Dispersion of Two Branches

Problem statement

Monthly revenues (thousand euros) of two bakery branches over the last $n = 6$ months:

Month	1	2	3	4	5	6
City branch	120	130	125	135	115	125
Rural branch	20	24	22	26	18	22

a) Compute the mean $\bar{x}$ and the sample variance s^2 for both branches.

b) Compute the standard deviation s.

c) Compute the coefficient of variation v and the normalised coefficient of variation v^*.

d) Which branch has the stronger revenue fluctuations? Which measure is suitable for this comparison, and why?

Solution (1/3) – Comparing the Dispersion of Two Branches

Rule

$s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$: square the deviations from the mean, add them up, divide by $n - 1 = 5$.

a) Means: $\bar{x}_{\text{city}} = \frac{750}{6} = \mathbf{125}$, $\bar{x}_{\text{rural}} = \frac{132}{6} = \mathbf{22}$ (check: the deviations sum to 0)

Month i	1	2	3	4	5	6	Total
City: $x_i - 125$	-5	5	0	10	-10	0	0
City: $(x_i - 125)^2$	25	25	0	100	100	0	250
Rural: $x_i - 22$	-2	2	0	4	-4	0	0
Rural: $(x_i - 22)^2$	4	4	0	16	16	0	40

$$s^2_{\text{city}} = \frac{250}{5} = \mathbf{50}, \qquad s^2_{\text{rural}} = \frac{40}{5} = \mathbf{8}$$

Solution (2/3) – Comparing the Dispersion of Two Branches

b) Standard deviation $s = +\sqrt{s^2}$:

$$s_{\text{city}} = \sqrt{50} = \mathbf{7.0711}, \qquad s_{\text{rural}} = \sqrt{8} = \mathbf{2.8284} \quad \text{(both in thousand euros)}$$

c) Coefficient of variation $v = s/\bar{x}$ and normalised coefficient of variation $v^* = v/\sqrt{n}$ with $\sqrt{6} = 2.4495$:

$$v_{\text{city}} = \frac{7.0711}{125} = \mathbf{0.0566} \qquad v^*_{\text{city}} = \frac{0.0566}{2.4495} = \mathbf{0.0231}$$

$$v_{\text{rural}} = \frac{2.8284}{22} = \mathbf{0.1286} \qquad v^*_{\text{rural}} = \frac{0.1286}{2.4495} = \mathbf{0.0525}$$

Observation

In absolute terms the city branch fluctuates more ($s = 7.07 > 2.83$), relative to the revenue level the rural branch does ($v = 12.9\,\% > 5.7\,\%$). Both v^* are close to 0 on the scale from 0 to 1 – and since n is equal, v^* gives the same ranking as v.

Solution (3/3) – Comparing the Dispersion of Two Branches

d) The branches operate on a **different scale** ($\bar{x} = 125$ vs. 22), so the **relative** measure v is suitable: $v_{\text{rural}} = 0.1286 > v_{\text{city}} = 0.0566 \Rightarrow$ **the rural branch has the stronger (relative) revenue fluctuations.**

```
city  = np.array([120, 130, 125, 135, 115, 125])
rural = np.array([20, 24, 22, 26, 18, 22])
for x in (city, rural):
    s = x.std(ddof=1)
    print(x.mean(), x.var(ddof=1), round(s, 4), round(s / x.mean(), 4))
```

```
125.0 50.0 7.0711 0.0566
22.0 8.0 2.8284 0.1286
```

Interpretation

$s = 7.07$ at a level of 125 weighs less than $s = 2.83$ at 22: the rural branch fluctuates by almost 13 % of its level, the city branch by less than 6 % – harder to plan for.

▶ **Try it yourself:** notebook, section 2.6 "Exercises" – open in Colab

Exam-Style Exercise – Books per Order

Problem statement

An online bookshop analyses the number of books per order for $n = 25$ orders:

Books per order x_j	1	2	3	4	5
Number of orders h_j	3	5	9	5	3

a) Complete the frequency table with f_j, H_j and $F(x_j)$.
b) Which share of the orders contains at most 2 books? Which share at least 4 books?
c) Determine the mode, the median, Q_1, Q_3 (use H_j) and the IQR.
d) Level of measurement of "number of books"? Why is the mean meaningful here – unlike for star ratings?
e) Compute the arithmetic mean from the frequency table.
f) Compute s^2, s and v from the frequency table.
g) Compare mean, median and mode: what do they say about the shape of the distribution?

Solution (1/4) – Books per Order

Rule

$f_j = h_j/n$; H_j and $F(x_j)$: add up the h_j or the f_j from the left; "at least 4" $= 1 - F(3)$.

a) Frequency table ($n = 25$; check: $H_5 = 25$, $F(5) = 1$):

Books x_j	1	2	3	4	5	Total
h_j	3	5	9	5	3	25
$f_j = h_j/25$	0.12	0.20	0.36	0.20	0.12	1
H_j	3	8	17	22	25	
$F(x_j)$	0.12	**0.32**	0.68	0.88	1	

b) At most 2 books: $F(2) = \frac{3+5}{25} = \mathbf{0.32}$ (32 % of the orders).
At least 4 books: $1 - F(3) = 1 - 0.68 = \mathbf{0.32}$ $(= \frac{5+3}{25}$, also 32 %).

Solution (2/4) – Books per Order

c) Mode: largest frequency $h_3 = 9 \rightarrow$ **3** books.

Quantiles: position $m = \lceil n \cdot p \rceil$; the value at position m is the first x_j with $H_j \geq m$.

	p	$n \cdot p$	Position	Value
Q_1	0.25	6.25	7	$H_1 = 3 < 7 \leq H_2 = 8 \rightarrow$ **2** books
Median	0.50	12.5	13	$H_2 = 8 < 13 \leq H_3 = 17 \rightarrow$ **3** books
Q_3	0.75	18.75	19	$H_3 = 17 < 19 \leq H_4 = 22 \rightarrow$ **4** books

$\text{IQR} = Q_3 - Q_1 = 4 - 2 =$ **2** books.

d) A count is **metric (ratio scale)** and discrete: natural zero, equal steps (1 to 2 books = 4 to 5 books), 4 books are twice as many as 2 $\rightarrow$ averaging is meaningful. Star ratings are only **ordinal**: the step from 1 to 2 stars need not equal that from 4 to 5, so "3.6 stars on average" has no exact meaning.

Solution (3/4) – Books per Order

Rule

$\bar{x} = \frac{1}{n}\sum_j h_j \cdot x_j$; $\quad s^2 = \frac{1}{n-1}\sum_j h_j \cdot (x_j - \bar{x})^2$; $\quad v = s/\bar{x}$

Books x_j	1	2	3	4	5	Total
h_j	3	5	9	5	3	25
$h_j \cdot x_j$	3	10	27	20	15	75
$h_j \cdot (x_j - 3)^2$	12	5	0	5	12	34

e) $\bar{x} = \frac{75}{25} = \mathbf{3}$ books

f) $s^2 = \frac{34}{24} = \mathbf{1.4167}$ (divide by $n - 1 = 24$), $\quad s = \sqrt{1.4167} = \mathbf{1.1902}$ books,
$v = \frac{1.1902}{3} = \mathbf{0.3967}$ (the standard deviation is about 40 % of the mean)

g) Mean = median = mode = 3; the frequencies $3, 5, 9, 5, 3$ mirror around $3 \rightarrow$ **symmetric** distribution (unlike the salaries, where an outlier pulls the mean above the median).

Solution (4/4) – Books per Order

Python check: `np.repeat` expands the table into the 25 raw values.

```
books = np.repeat([1, 2, 3, 4, 5], [3, 5, 9, 5, 3])        # expand the table
print(len(books), (books <= 2).mean(), (books >= 4).mean())
print(pd.Series(books).mode().tolist(),
      np.quantile(books, [0.25, 0.5, 0.75], method="inverted_cdf"))
s = books.std(ddof=1)
print(books.mean(), round(s**2, 4), round(s, 4), round(s / books.mean(), 4))
```

```
25 0.32 0.32
[3] [2 3 4]
3.0 1.4167 1.1902 0.3967
```

Key takeaway

Exam routine: frequency table first (f_j, H_j, F) – then read off shares and quantiles, and compute $\bar{x}$, s^2, s, v with the weights h_j.

▶ **Try it yourself:** notebook, section 2.7 "Exam-Style Exercise" – open in Colab

Overview

1 Fundamentals

2 Descriptive Statistics

3 Summary

Summary 1: Key Concepts

Fundamentals

- Statistical unit (ω): object being observed, delimited by subject, time, location
- Variable (X): characteristic of interest; Observation (x_i): actually observed value
- Levels of measurement: nominal (categories) $\rightarrow$ ordinal (ranking) $\rightarrow$ metric (differences meaningful; interval scale, or ratio scale with a natural zero)
- Metric data is required for most statistical analyses in this course

Frequencies

- $h_j = \#\{i \mid x_i = x_j\}, \quad f_j = h_j/n, \quad H_j = \sum_{l \leq j} h_l, \quad F(x) = \sum_{x_j \leq x} f_j$
- "At most x" $= F(x)$; "more than x" $= 1 - F(x)$
- Percentage points = absolute difference of two shares; percent = relative change

Summary 2: Measures of Central Tendency

Measure	Formula	Properties
Arithmetic mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{1}{n}\sum_{j=1}^{k} h_j x_j$	Easy to compute; sensitive to outliers
Trimmed mean	Mean without $\lfloor \alpha n \rfloor$ values per end	Robust to outliers
Geometric mean	$\bar{x}_{\text{geo}} = \left(\prod_{i=1}^{n} x_i\right)^{1/n}$	Growth factors: rate $= \bar{x}_{\text{geo}} - 1$
Median	$\tilde{x} = x_{(\lceil 0.5 \cdot n \rceil)}$	Robust to outliers; also ordinal data
Mode	Most frequent value	Also nominal data; often not unique

Quantiles

For the p-quantile: position $= \lceil n \cdot p \rceil$ in the sorted data (no rounding if $n \cdot p$ is an integer).
Quartiles: $Q_1 = x_{(\lceil 0.25n \rceil)}$, $Q_2 = x_{(\lceil 0.5n \rceil)}$ (Median), $Q_3 = x_{(\lceil 0.75n \rceil)}$

Summary 3: Measures of Dispersion

Measure	Formula	Key property
Range R	$x_{(n)} - x_{(1)}$	Determined by the two extreme values
Interquartile range	$Q_3 - Q_1$	Width of the middle 50 %; robust
Variance s^2	$\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$	Dimension changes (e.g. cm^2)
Std. deviation s	$+\sqrt{s^2}$	Same unit as original data
Coeff. of variation v	$s/\bar{x}$	Scale-independent (positive data)
Norm. coeff. v^*	$v/\sqrt{n}$	Normalised to $[0, 1]$; decreases with n

Which measure for which comparison?

- Same scale: s^2 or s sufficient (n only affects the precision)
- Different scale: use v
- v^*: rates the relative dispersion between 0 and 1; **not** for different n

Summary 4: Key Python Commands (1)

Frequencies

```
data.value_counts().sort_index()                      # Absolute frequencies
data.value_counts(normalize=True).sort_index()        # Relative frequencies
data.value_counts(normalize=True).sort_index().cumsum()  # Cumulative
```

Measures of Central Tendency

```
np.mean(data)                                  # Arithmetic mean
stats.trim_mean(data, 0.10)                    # Trimmed mean (10% per end)
stats.gmean(factors) - 1                       # Average growth rate
np.median(data)                                # Median (even n: averages!)
data.mode()                                    # Mode
np.quantile(data, 0.25, method="inverted_cdf") # Quantiles (e.g. Q1)
```

Summary 4: Key Python Commands (2)

Imports and frequency tables

```
import numpy as np
import pandas as pd
from scipy import stats
np.average(values, weights=counts)      # Mean from a frequency table
np.repeat(values, counts)               # Frequency table -> raw data
```

Measures of Dispersion

```
np.ptp(data)                                    # Range
q1, q3 = np.quantile(data, [0.25, 0.75], method="inverted_cdf")
q3 - q1                                         # Interquartile range
np.var(data, ddof=1)                            # Sample variance (ddof=1!)
np.std(data, ddof=1)                            # Standard deviation
np.std(data, ddof=1) / np.mean(data)            # Coefficient of variation
np.std(data, ddof=1) / np.mean(data) / np.sqrt(len(data))  # Norm. coeff.
```

▶ **Try it yourself:** notebook, section 3 "Summary" (key Python commands) – open in Colab

Outlook: Part 2 – Graphical Representations

What we have so far

The shoe sizes are now described by **numbers**: frequencies, measures of central tendency and measures of dispersion.

Next part

The same data are described by **pictures**: the empirical cumulative distribution function, the histogram and the boxplot – the quartiles and the IQR of this part reappear as the building blocks of the boxplot.

Key takeaway

Numbers and pictures complement each other: a good description of data always uses both.